

(Lecture 22)

Tailwind CSS CLI - Complete Installation Guide

Course: PW EARNERS - Coding and Web Development - **For:** Mac aur Windows dono

Yeh guide step-by-step hai - har step ka output dikhaya gaya hai taaki tum compare kar sako ki tumhare screen pe kya aa raha hai. Agar kuch alag dikhe - us section mein diya hua fix follow karo.

Before We Start - Kya Chahiye?

Tailwind CLI ke liye **Node.js** aur **npm** chahiye. Pehle check karte hain ki installed hain ya nahi.

Step 1 - Terminal / Command Prompt Kholo

Mac pe

- **Cmd + Space** dabao → "Terminal" type karo → Enter
- Ya: Applications → Utilities → Terminal

Windows pe:

- **Windows key** dabao → "cmd" type karo → **Command Prompt** pe click karo
- Ya: Start Menu → Windows System → Command Prompt

Tip: VS Code mein bhi terminal open kar sakte ho - **Ctrl + `** (backtick) press karo. Yeh sabse easy hai kyunki project bhi wahi open hoga.

Step 2 - Node.js Check Karo

Terminal mein yeh type karo aur Enter dabao:

```
node -v
```

Agar yeh dikhe (version number) - Node installed hai:

```
v20.11.0
```

(version number alag ho sakta hai - v18 ya usse upar koi bhi version theek hai)

Step 3 pe jao directly.

Agar yeh dikhe - Node installed NAHI hai:

Mac pe:

zsh: command not found: node

Windows pe:

'node' is not recognized as an internal or external command

Node install karo - neeche diya hai.

Step 2A - Node.js Install Karo (Sirf agar Step 2 mein error aaya)

Mac pe Node Install Karna:

Option 1 - Official website se (Recommended):

1. Browser mein jao: <https://nodejs.org>
2. **"LTS"** wala button click karo (Long Term Support - stable version)
3. **.pkg** file download hogi
4. Download hone ke baad file open karo - installer chalega
5. "Continue", "Agree", "Install" karte jao
6. Install complete hone ke baad terminal **band karo aur dobara kholo**

Option 2 - Homebrew se (agar Homebrew already installed hai):

brew install node

Windows pe Node Install Karna:

1. Browser mein jao: <https://nodejs.org>

2. **"LTS"** wala button click karo
3. **.msi** file download hogi
4. Download hone ke baad file open karo - installer chalega
5. "Next", "I Accept", "Next", "Next", "Install" karte jao
6. Ek jagah poochha jayega **"Automatically install the necessary tools"** - yeh **check karo** (tick karo)
7. Install complete hone ke baad **Command Prompt band karo aur دوبارا kholo**

Important - Windows: Install ke baad Command Prompt **restart karna zaroori hai** - warna naya command recognize nahi hoga.

Step 3 - Node Version Dobara Check Karo

Terminal mein دوبارا type karo:

```
node -v
```

Ab yeh dikhna chahiye:

```
v20.11.0
```

(v18 ya usse upar koi bhi version theek hai)

Abhi bhi error aa raha hai?

Mac pe:

- Terminal band karo, دوبارا kholo, phir try karo
- Agar phir bhi error - website se download karke installer دوبارا chalao

Windows pe:

- Command Prompt band karo, دوبارا kholo, phir try karo
 - Agar phir bhi - Computer restart karo, phir try karo
 - Agar abhi bhi - installer mein **PATH** option check hua tha? Reinstall karo aur dhyan se check karo
-

Step 4 - npm Check Karo

npm -v

Agar version dikhé - npm installed hai:

10.2.4

(version number alag ho sakta hai - 8 ya usse upar theek hai)

Step 5 pe jao.

Agar error aaye:

npm Node.js ke saath aata hai - agar Node install hua hai aur npm nahi chal raha:

Mac pe:

sudo npm install -g npm@latest

(password maangega - apna Mac password daalo, typing nahi dikhegi - normal hai)

Windows pe:

npm install -g npm@latest

Phir dobara check karo: `npm -v`

Step 5 - Project Folder Banao

Yeh ek naya folder hai jahan Tailwind project rahega.

Mac pe:

cd Desktop

mkdir tailwind

cd tailwind

Windows pe:

cd Desktop

mkdir tailwind

cd tailwind

VS Code mein folder kholo:

code .

(. ka matlab current folder)

Yeh dikhna chahiye:

Mac pe:

/Users/yourname/Desktop/tailwind

Windows pe:

C:\Users\yourname\Desktop\tailwind

Step 6 - npm Project Initialize Karo

npm init -y

Yeh dikhna chahiye:

Wrote to /Users/yourname/Desktop/tailwind/package.json:

```
{  
  
  "name": "tailwind",  
  
  "version": "1.0.0",  
  
  "description": "",  
  
  "main": "index.js",
```

```
"scripts": {  
  
  "test": "echo \"Error: no test specified\" && exit 1"  
  
},  
  
"keywords": [],  
  
"author": "",  
  
"license": "ISC"  
}
```

- "-y" flag ka matlab - sab default accept karo, koi sawaal mat poochho

Folder mein ek `package.json` file ban gayi. VS Code mein check kar sakte ho.

Step 7 - Tailwind CSS Install Karo

`npm install tailwindcss @tailwindcss/cli`

Yeh dikhna chahiye (thoda time lagega - 30 seconds to 2 minutes):

added 113 packages, and audited 114 packages in 45s

17 packages are looking for funding

run ``npm fund`` for details

found 0 vulnerabilities

(numbers alag ho sakte hain - theek hai)

Agar yeh error aaye:

"ENOTFOUND" ya network error:

npm error code ENOTFOUND

npm error errno ENOTFOUND

Internet connection check karo. WiFi se connected ho? Try again.

"EACCES" error (Mac):

npm error code EACCES

npm error errno -13

Permission error hai. Yeh try karo:

sudo npm install tailwindcss @tailwindcss/cli

"EACCES" error (Windows): ➡ Command Prompt ko **Administrator** mode mein kholo:

- Start Menu → "cmd" search karo → Right click → "Run as Administrator"
- Phir dobara navigate karo folder mein aur install karo

Step 8 - Project Files Banao

File 1 - **input.css** banao

VS Code mein ya terminal se:

Terminal se (Mac):

touch input.css

Terminal se (Windows):

type nul > input.css

Ya VS Code mein: New File button → naam **input.css**

Ab **input.css** kholo aur yeh likho:

```
@import "tailwindcss";
```

Bas - sirf yeh ek line.

File 2 - `index.html` banao

Terminal se:

Mac:

`touch index.html`

Windows:

`type nul > index.html`

Ab `index.html` kholo aur yeh likho:

```
<!doctype html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Tailwind</title>
```

```
  <link rel="stylesheet" href="output.css">
```

```
</head>
```

```
<body class="bg-gray-100 min-h-screen p-8">
```

```
  <h1 class="text-4xl font-bold text-gray-900 mb-4">
```

```
    Tailwind
```

```
  </h1>
```

```
  <p class="text-gray-600 text-lg mb-6">
```

Wear the Code. Live the Street.

</p>

<button class="bg-black text-white px-6 py-3 rounded-lg font-semibold hover:bg-red-500 transition-colors">

Shop Now

</button>

</body>

</html>

Step 9 - Tailwind Build Karo (Watch Mode)

`npx @tailwindcss/cli -i input.css -o output.css --watch`

Yeh dikhna chahiye:

Rebuilding...

Done in 236ms

- Yeh **watch mode** hai - jab bhi tum koi file save karo, Tailwind automatically rebuild karega
- Terminal **band mat karo** jab tak kaam kar rahe ho - background mein chalta rehna chahiye
- Ek nayi file **output.css** ban gayi hogi folder mein - yeh Tailwind ka generated CSS hai

Agar yeh error aaye:

"Cannot find module" error:

Error: Cannot find module 'tailwindcss'

Tailwind sahi install nahi hua. Phir se karo:

`npm install tailwindcss @tailwindcss/cli`

"npx" not found (Windows):

'npx' is not recognized as an internal or external command

Node.js sahi install nahi hua ya PATH set nahi hua. Node.js reinstall karo - installer mein "Add to PATH" option zaroor check karo.

Step 10 - Result Dekho

VS Code mein **Live Server** extension se `index.html` kholo:

1. `index.html` pe right click karo
2. "Open with Live Server" click karo
3. Browser mein page khulega

Yeh dikhna chahiye:

- Gray background page
- "Tailwind" - bada bold heading
- Gray colored tagline text
- Black button jo hover pe red ho jaata hai

Agar styles nahi dikh rahe:

- `output.css` file exist karti hai folder mein?
 - `index.html` mein `<link rel="stylesheet" href="output.css">` sahi likha hai?
 - Terminal mein watch mode chal raha hai? (Done in XXXms dikhna chahiye)
-

Project Structure - Ab Yeh Hona Chahiye

tailwind/

|

|— index.html ← HTML file (tum edit karte ho)

|— input.css ← Tailwind import (tum likhte ho)

|— output.css ← Generated CSS (auto - touch mat karo)

|— package.json ← npm project info

|— node_modules/ ← Installed packages (touch mat karo)

Note: `node_modules` folder bahut bada hoga (100MB+) - yeh normal hai. Isko kabhi delete mat karo manually. Agar koi bhi issue aaye - `npm install` run karo, woh wapas rebuild ho jaayega.

Step 11 - Daily Workflow (Har Din Yeh Karo)

Jab bhi project pe kaam karna ho:

Step 1 - Terminal kholo project folder mein:

```
cd Desktop/tailwind
```

Step 2 - Watch mode start karo:

```
npx @tailwindcss/cli -i input.css -o output.css --watch
```

Step 3 - VS Code mein files edit karo, browser mein Live Server pe dekho

Step 4 - Kaam khatam hone pe terminal mein **Ctrl + C** dabao watch band karne ke liye

`package.json` mein Shortcut Script Add Karo (Optional but Helpful)

Baar baar `npx @tailwindcss/cli -i input.css -o output.css --watch` type karna boring hai.

`package.json` kholo aur "`scripts`" section mein yeh add karo:

```
{  
  
  "name": "tailwind",
```

```

"version": "1.0.0",

"scripts": {

  "dev": "npx @tailwindcss/cli -i input.css -o output.css --watch",

  "build": "npx @tailwindcss/cli -i input.css -o output.css --minify"

},

"dependencies": {

  "tailwindcss": "^4.0.0",

  "@tailwindcss/cli": "^4.0.0"

}

}

```

Ab sirf yeh type karo watch mode ke liye:

```
npm run dev
```

Aur production build ke liye (minified, chhoti file):

```
npm run build
```

Common Errors - Quick Reference

Error	Cause	Fix
node: command not found	Node install nahi hai	nodejs.org se install karo
npm: command not found	Node install nahi hua properly	Node reinstall karo
ENOTFOUND	Internet issue	Connection check karo, retry
EACCES (Mac)	Permission issue	sudo npm install use karo

Error	Cause	Fix
Cannot find module	npm install incomplete	npm install dobara run karo
npx not recognized (Windows)	PATH set nahi	Node reinstall, "Add to PATH" check karo
Styles nahi dikh rahe	output.css missing ya link galat	Watch mode chala hai? Link sahi hai?
output.css empty	input.css mein @import "tailwindcss" missing	input.css check karo

Versions Check - Summary Table

Yeh table students ke liye - class mein check karo sabka:

Check	Command	Expected Output
Node version	node -v	v18.x.x ya usse upar
npm version	npm -v	8.x.x ya usse upar
Tailwind installed?	npx @tailwindcss/cli --version	4.x.x

Quick Commands Reference

Node check

node -v

npm check

npm -v

New project initialize

npm init -y

Tailwind install

npm install tailwindcss @tailwindcss/cli

Watch mode start

npx @tailwindcss/cli -i input.css -o output.css --watch

Watch mode (agar script add ki hai)

npm run dev

Production build

npm run build

Watch band karna

Ctrl + C